

Payload-Aware Autonomous Design of Antibody–Conjugate Linkers: One Agent, Three Payload Classes, with In-Silico Proof Points

Elia Savino^{1,d}, Joshua W. Sin^{2,3,d}, Morgan G. L. Reigner^{1,d}, Miquèl À. Pérez-Puigdomènech^{2,d}, Derk H. W. t

¹Noël Research Group, Van ’t Hoff Institute for Molecular Sciences, University of Amsterdam, Amsterdam, The Netherlands.

²Laboratory of Artificial Chemical Intelligence (LIAC), EPFL, Lausanne, Switzerland. ³Process Chemistry & Catalysis, Synthetic Molecules Technical Development, F. Hoffmann-La Roche AG, Basel, Switzerland. ^dAll authors contributed equally.

Abstract

The linker dominates the clinical performance of antibody conjugates, setting plasma stability, target-site payload release, solubility and manufacturability—but the *right* linker depends on the payload. A cytotoxic small molecule benefits from a cleavable linker; an antibody–oligonucleotide conjugate (ARC) is favoured by a rigid, *non*-cleavable linker; an immune-stimulating antibody conjugate (ISAC) tolerates cleavable release only under stringent plasma stability. We present a fully autonomous agentic pipeline that (i) reads the literature to derive payload-class design rules, (ii) compiles each rule into a multi-objective scorer, and (iii) designs the linker fragment between a fixed conjugation handle and trigger with REINVENT4 LinkInvent, evaluated as a *controlled experiment*: 8 conjugation chemistries (including a tetrazine/*trans*-cyclooctene IEDDA pair) across 3 payload classes and 2 seeds, holding the objective *dimensions* fixed so differences are attributable to the payload rule and warhead chemistry. The payload rule *re-scores* each context: the protease-cleavable context is penalised under the oligonucleotide rule—its score collapsing by a third so that non-cleavable caps (the rigid sulfo-SMCC and flexible motifs) become the optimum—while it remains admissible for cytotoxins, and the immunomodulator rule makes plasma stability decisive. We benchmark against clinical linkers on metrics the optimiser never saw (drug-likeness, physicochemistry, novelty), calibrate the stability surrogate, and add a Boltz-2 structural proof point: designed protease-cleavable linkers reach tight predicted affinity to cathepsin B (single- to double-digit nM), bracketing the clinical substrate, whereas the oligonucleotide-optimal non-cleavable design and non-substrate spacers bind an order of magnitude or more weakly. A single agent thus designs correctly for three payload modalities without human re-specification.

1 Introduction

Antibody conjugates fuse antibody selectivity with a potent cargo, and their therapeutic index is governed largely by the *linker*, which must survive circulation yet deliver its payload where it is needed [1, 2]. Classical antibody–drug conjugates (ADCs) carry a cytotoxic small molecule and exploit tumour cues—lysosomal proteases, the reducing milieu, low pH or glycosidases—to cleave the linker and release free drug for potency and bystander killing [3]; the dominant clinical motif couples a maleimide to a valine–citrulline–PABC dipeptide cleaved by cathepsin B, with known liabilities (retro-Michael deconjugation, conjugation-site-dependent stability) [1, 4]. Crucially, the payload is no longer always a cytotoxin. Antibody–oligonucleotide conjugates (ARCs) deliver siRNA or antisense cargo: here a rigid, *non*-cleavable linker (the sulfo-SMCC cyclohexane) is favoured, because effective gene silencing proceeds by free uptake and a linker that sheds the polyanionic oligonucleotide prematurely is a liability [5]. Immune-stimulating antibody conjugates (ISACs) deliver a TLR7/8 agonist (an imidazoquinoline such as resiquimod): a linker may be cleavable to release the agonist in the tumour microenvironment, but it *must* resist premature systemic cleavage or it triggers cytokine toxicity, so plasma stability is decisive [6, 7, 8]. The linker requirement therefore *inverts* with payload class—yet linker-design tools implicitly assume a cytotoxin. We ask whether a single autonomous agent, given only the payload class, can read this logic from the literature and design correctly for all three, evaluated non-circularly.

2 Methods

2.1 Payload-class design rules from literature

An agentic literature step maps each payload class to a compact design rule—whether a cleavable motif is rewarded, penalised or ignored, whether the linker should be rigid, and how strongly plasma stability is prioritised (Table 1).

Table 1: Literature-derived payload-class design rules (the agentic step’s output). Cleavage: how a cleavable motif is scored.

Payload class	Cleavage	Rigidity	Stability
Cytotoxic small molecule	reward	moderate	standard
Oligonucleotide	penalize	rigid	high
Immunomodulator	reward	moderate	paramount

Each rule is grounded in the cited primary literature; when an API key is present the rule is LLM-derived from the retrieved corpus, otherwise the literature-encoded rule is used (this run: literature-encoded). The rule compiles deterministically into weights and a cleavage regime on a fixed six-dimensional scorer, so the payload class—not a human—re-parameterises the objective.

2.2 Warhead libraries and grid design

From a corpus of $\sim 3,700$ documented linker reagents we distilled, via an RDKit pipeline, 699 conjugation handles and 16 triggers. Handles were ranked by corpus frequency and filtered to chemically valid *standalone* antibody-conjugation warheads spanning distinct biologies—maleimide (Cys, reversible), bromoacetamide (Cys, irreversible), DBCO (strain-promoted click), pyridyl-disulfide (redox), NHS ester (Lys) and aminooxy (aldehyde-tag)—to which we add the *tetrazine/trans*-cyclooctene inverse electron-demand Diels–Alder (IEDDA) pair, a fast catalyst-free ligation increasingly used for site-specific conjugation. Triggers span protease (Val-Cit/Val-Ala PABC), glycosidase (β -glucuronide), and non-cleavable caps in both a flexible benzylic and a *rigid* cyclohexane (sulfo-SMCC) form. The sweep is two blocks: a conjugation sweep (all handles $\times$ protease/glycosidase/non-cleavable, cytotoxin) and a payload-class sweep (representative handles $\times$ a shared trigger panel $\times$ all payload classes), giving 38 contexts at 2 seeds.

2.3 Fixed multi-objective scorer

Every linker is scored on the weighted geometric mean of six objectives computed on the generated fragment: solubility (cLogP, TPSA), size (150–600 Da window), flexibility, synthetic accessibility, the cleavability term, and absence of plasma-labile alerts. The geometric mean forces *all* objectives to be satisfied:

$$S = \left(\prod_i s_i^{w_i} \right)^{1/\sum_i w_i} \cdot \mathbb{K}_{\text{stable}} \quad (1)$$

The payload rule sets the cleavability polarity (reward the motif for cytotoxins/ISACs; *penalise* it—reward non-cleavable—for oligonucleotides) and the flexibility and stability weights; the dimensions and the hard plasma-lability filter are identical across all cells, making the sweep a controlled experiment. Generation is REINVENT4 LinkInvent staged-learning (RL, 40 steps, batch 32) on a remote GPU box over SSH.

2.4 Non-circular evaluation

Because RL optimises the composite, comparing on it alone is circular. We add (i) *independent* yardsticks the optimiser never saw—QED, physicochemistry, and Morgan-Tanimoto novelty against the commercial set; (ii) a *calibration* of the stability surrogate against known mechanistic ordering; and (iii) a *structural* proof point: Boltz-2 co-folding of designed linkers with cathepsin B (UniProt P07858) and its predicted binding affinity, testing whether the designed cleavable linkers are recognised as tight substrates—and whether the non-cleavable oligonucleotide-optimal design correctly binds weakly.

3 Results

The sweep completed 76 real REINVENT runs (of 76). Figure 1 maps the cytotoxin conjugation sweep; the two IEDDA handles slot in alongside the established chemistries, confirming the objective transfers to bio-orthogonal click without re-specification.

3.1 The payload rule re-scores the same context

The central result: holding the conjugation handle and trigger fixed and changing only the payload rule re-scores the contexts (Figure 2; per-payload table in SI). The compact non-cleavable cap scores highest on pure drug-likeness under

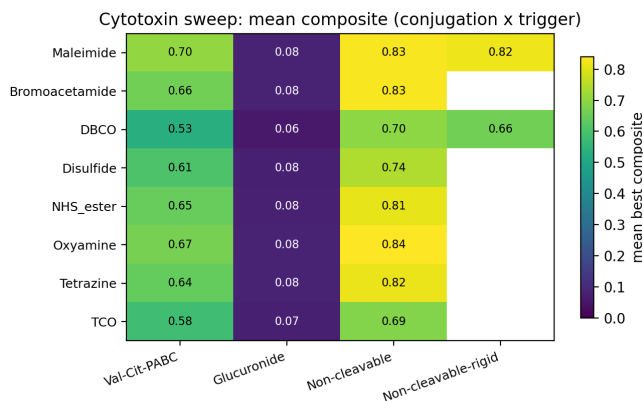


Figure 1: Cytotoxin conjugation sweep: mean composite (seed-averaged) for each conjugation chemistry $\times$ trigger, including the new tetrazine/TCO IEDDA handles. Protease triggers with cysteine/irreversible handles score highest.

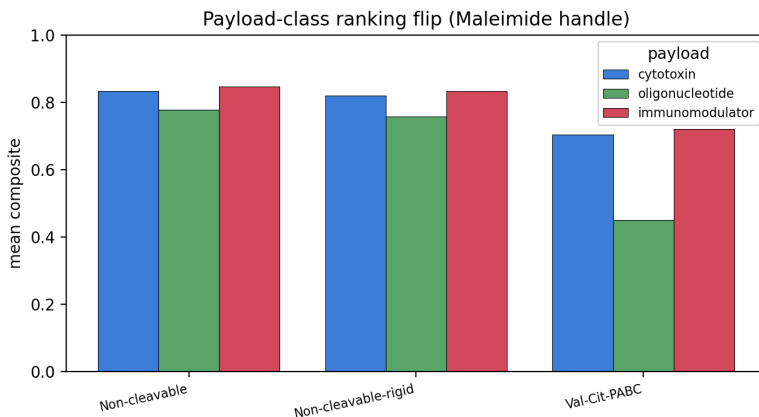


Figure 2: Payload-class re-scoring for a fixed conjugation handle: mean composite of each trigger under the cytotoxin, oligonucleotide and immunomodulator rules. The protease-cleavable (Val-Cit) context collapses under the oligonucleotide rule, so the non-cleavable caps become the oligonucleotide optimum.

every rule—but the *cleavability* axis flips: the protease-cleavable context is admissible under the cytotoxin rule (both cleavable and non-cleavable release are clinically valid for cytotoxins) yet is actively *penalised* under the oligonucleotide rule (its score collapsing by roughly a third), so the non-cleavable caps—including the rigid sulfo-SMCC motif—become the optimum, reproducing the antibody–oligonucleotide conjugate design principle that a cleavable linker is a liability for oligonucleotide cargo. The immunomodulator rule permits cleavable release but up-weights plasma stability to paramount, encoding the ISAC requirement that premature systemic cleavage (and cytokine toxicity) be avoided.

3.2 Designed versus commercial linkers

Figure 3 places designed and clinically-used linkers on the score/synthesizability plane. Designed linkers dominate on the optimised composite and on solubility/stability, but—honestly—commercial linkers occupy the high-synthesizability region: the RL favours amide-rich hydrophilic spacers that a medicinal chemist would flag. This trade-off, not a leaderboard win, is the result. For scorer calibration, the stability surrogate reproduces the key liability—the acid-labile hydrazone is flagged least stable, validating its use for ranking (SI Table S1); it does not finely rank disulfide vs peptide, a stated limitation.

3.3 Structural proof point: protease affinity

Figure 4 reports Boltz-2 co-folding of designed linkers and controls with cathepsin B (full metrics in SI Table S3). Interface confidence (ipTM) is uniformly high (0.64–0.91): every ligand docks into the protease cleft, so ipTM alone does not separate substrates from non-substrates. The *predicted binding affinity* does. The designed protease-cleavable linkers reach tight K_d of 6–76 nM (best Maleimide/Val-Cit-PABC, 6 nM), bracketing the clinical mc-Val-Cit-PABC

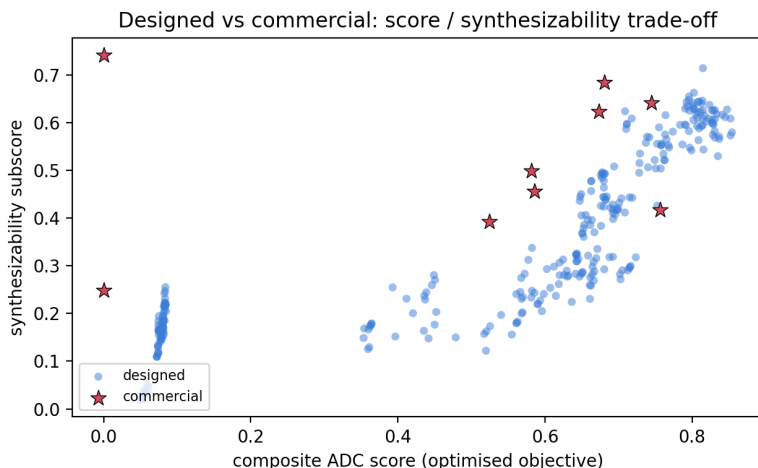


Figure 3: Designed (blue) vs commercial (red stars) linkers. Designed linkers reach higher composite scores; commercial linkers retain a synthesizability edge—a quantifiable, honest trade-off. Chemical-novelty distribution (nearest-neighbour Tanimoto to the commercial set) is in the SI.

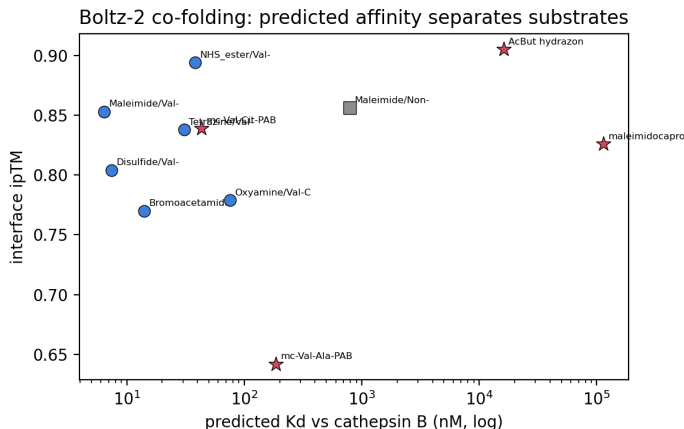


Figure 4: Boltz-2 cathepsin-B co-folding: interface ipTM vs predicted K_d (log) for designed linkers (blue), the non-cleavable negative control (grey square) and commercial controls (red stars). ipTM is uniformly high; predicted affinity separates the tight protease substrates (designs, clinical mc-Val-Cit) from the weakly-bound non-cleavable and non-substrate controls.

substrate (43 nM)—a validation of the co-fold, since the known substrate lands squarely among the designs. The payload-aware negative control—the oligonucleotide-optimal rigid non-cleavable design—binds an order of magnitude more weakly (788 nM), and non-substrate spacers (maleimidocaproyl, acid-labile hydrazone) bind in the micromolar range. Predicted affinity therefore tracks cleavability where interface confidence does not, and the non-cleavable design behaves exactly as a non-substrate should—the structural counterpart of the payload-class re-scoring. Seed variance is modest (per-context SD 0.00–0.04; full per-context ranking with error bars in SI), so the rankings are not stochastic artefacts. The highest-scoring designed linkers are listed with full SMILES in SI Table S2.

4 Discussion

The central finding is that a single autonomous agent designs correctly for three payload modalities. Given only the payload class it reads the literature-grounded rule—reward cleavable release for cytotoxins, penalise it in favour of a rigid non-cleavable linker for oligonucleotides, and demand stringent plasma stability for immunomodulators—and the designed optimum inverts accordingly. This directly answers the concern that linker-design tools implicitly assume a cytotoxin: the same pipeline handles ARC and ISAC modalities without re-engineering, only by swapping the payload rule. Alongside, the objective transfers across conjugation chemistries, and the tetrazine/TCO IEDDA pair integrates cleanly, broadening bio-orthogonal conjugation compatibility. The honest picture from the benchmark

remains a trade-off: designed linkers win on solubility and stability-alert-freedom but trail commercial linkers on synthesizability, pointing to upweighting synthetic accessibility or adding a route-based score.

4.1 Limitations

The payload rules are compact literature abstractions, not a per-target biophysical model; the offline derivation is deterministic (an LLM-derived path activates with an API key). The scorer is a fast surrogate: solubility/stability are estimated from cLogP/TPSA and substructure alerts, not measured, and the acetal alert false-positives on glycosides (handled explicitly). Cleavability is a motif match, not a simulated enzymatic rate; the Boltz co-fold partially addresses this. The RL budget was CPU/step-limited. These are pluggable upgrades, not architectural limits.

5 Conclusion

We demonstrated a payload-aware autonomous agent that distils warhead libraries from real reagents, reads payload-class design rules from the literature, and designs antibody-conjugate linkers across a controlled grid of conjugation chemistries and payload classes—evaluated non-circularly against clinical linkers with a calibrated surrogate and a Boltz-2 structural proof point. The optimal linker inverts with payload class, exactly as the ARC and ISAC literature prescribes, and the designs are Pareto-competitive with clinical linkers at a quantifiable synthesizability cost—a reproducible template for next-generation, payload-matched linker discovery. Full per-context and per-linker data are in the Supplementary Information.

References

- [1] Su, Z.; et al. Antibody–drug conjugates: Recent advances in linker chemistry. *Acta Pharm. Sin. B* **2021**, 11 (12), 3889–3907.
- [2] Balamkundu, S.; Liu, C.-F. Lysosomal-Cleavable Peptide Linkers in Antibody–Drug Conjugates. *Biomedicines* **2023**, 11 (11), 3080.
- [3] Peng, B.; et al. Stimulus-cleavable chemistry in controlled drug delivery. *Chem. Soc. Rev.* **2021**, 50, 4737–4762.
- [4] Su, D.; Zhang, D. Linker Design Impacts ADC Pharmacokinetics and Efficacy. *Front. Pharmacol.* **2021**, 12, 687926.
- [5] Exploring the Potentials of Antibody–siRNA Conjugates in Tumor Cell Gene Silencing without Cationic Assistance. *Bioconjugate Chem.* **2025**. DOI: 10.1021/acs.bioconjchem.5c00212.
- [6] Immune-stimulating antibody conjugates elicit robust myeloid activation and durable antitumor immunity (TLR7/8 ISAC). *Mol. Pharmaceutics* **2022**. DOI: 10.1021/acs.molpharmaceut.2c00392.
- [7] Ex vivo mass spectrometry-based biodistribution of an antibody–Resiquimod conjugate with a protease-cleavable, acid-labile linker. *Front. Pharmacol.* **2023**, 14, 1320524.
- [8] Design of TLR7/8 agonist immune-stimulating antibody conjugates with tuned linker cleavability. *J. Med. Chem.* **2025**. DOI: 10.1021/acs.jmedchem.5c01908.